# Transport Layer

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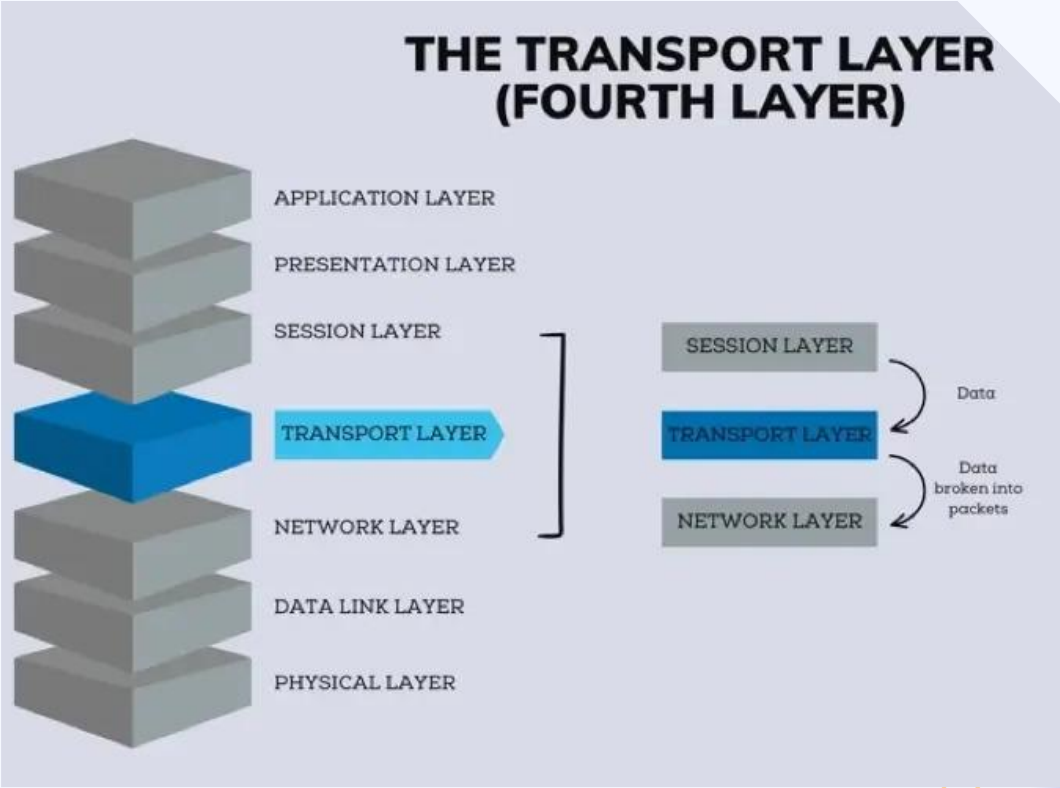
# The Transport layer

The Transport Layer is the 4th layer of the OSI model, and the 3rd layer of the TCP/IP model



# What is the transport layer

The Transport Layer is the 4th layer of the OSI model. It acts as a bridge between the upper layers (application, session, presentation) and the lower layers (network, data link, physical). Its primary role is to provide logical communication between application processes running on different hosts.

# Key Responsibilities

## Segmentation

Divides large messages into smaller segments and reassemble at location

## End-to-End

Manages complete data transfer from source to destination  application

## Port addressing

Identifies specific applications using the hosts port number

## Error Detection

Checks for corruption and retransmits lost or damaged data

## Connection Management

Establishing and terminating connections through the handshake method

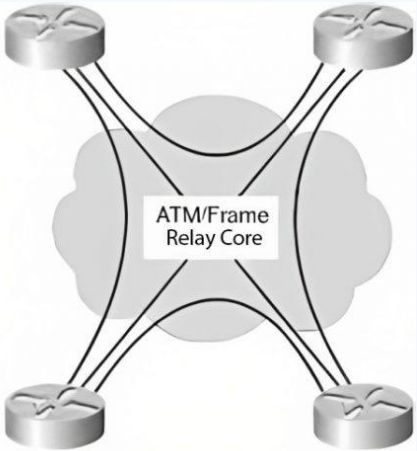
## Flow Control

Regulates data transmission speed to prevent network overload

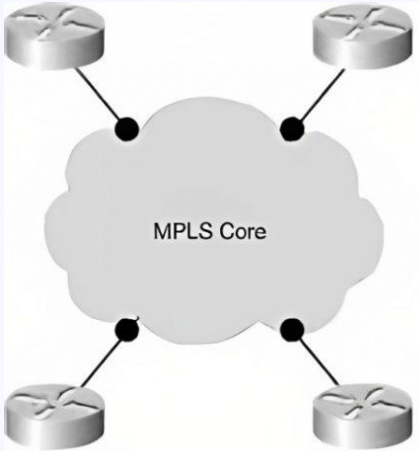
# TCP VS UDP

## TCP

- Transmission control Protocol
- Connection bassed
- Ordered
- Flow/congestion control
- Harder to implement



Connection-Oriented



Connectionless

## UDP

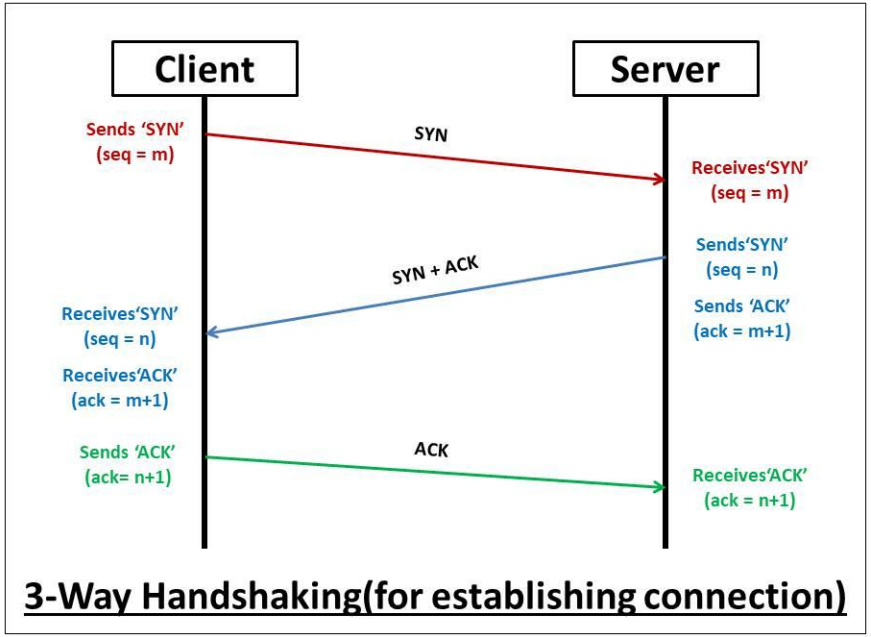
- User datagram protocol
- Conectionless
- No ordering
- No flow/congetsion control
- Easier to implement

# TCP in depth

- TCP treats data as a continuous stream of bytes.
- Before any data is sent, a connection must be established via the three-way handshake (SYN → SYN-ACK → ACK).
- Every byte sent is numbered with a sequence number, and the receiver must acknowledge receipt.
- If an acknowledgement isn't received within a timeout period, the data is retransmitted automatically.
- Key Mechanisms:
  - segmentation
  -sequence numbers
  -flow/congestion control
  -connection termination

# Three way handshake

How TCP establishes connections befor data transfer



# UDP in depth

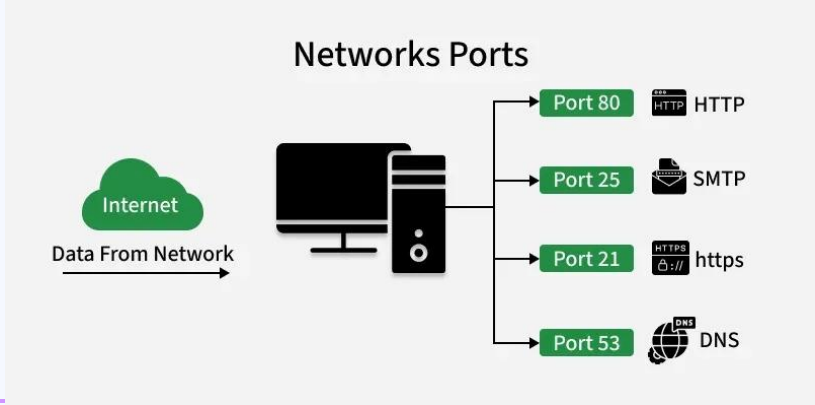
- UDP simply packages application data into a datagram and sends it.
- There is no handshake, no acknowledgement, no retransmission, and no ordering guarantee.
- The sender fires packets and forgets them.
- If a packet is lost, corrupted, or arrives out of order, UDP does nothing — it's entirely up to the application layer to handle this if needed.
- Key characteristics:
  - multicast & broadcast support
  -checksum

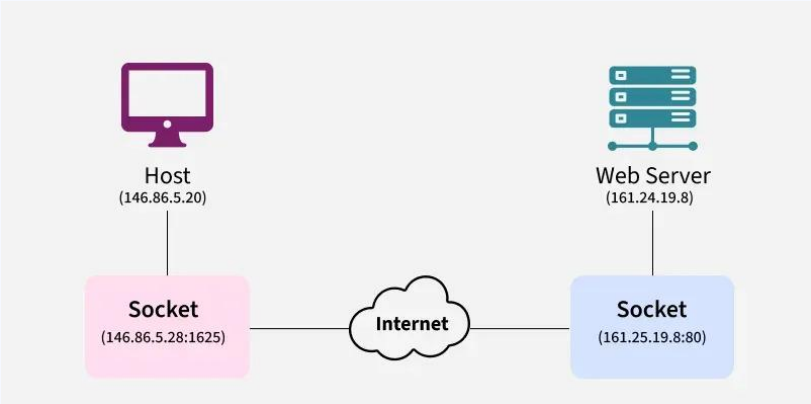
# Ports and Sockets

Port = 16 bit integer  that identifies a service
Its how the transport layer identifies which application should receive data

SOCKET = IP Address + Port Number

Key ideas: client socket, socket pair, server socket, and Multiplexing



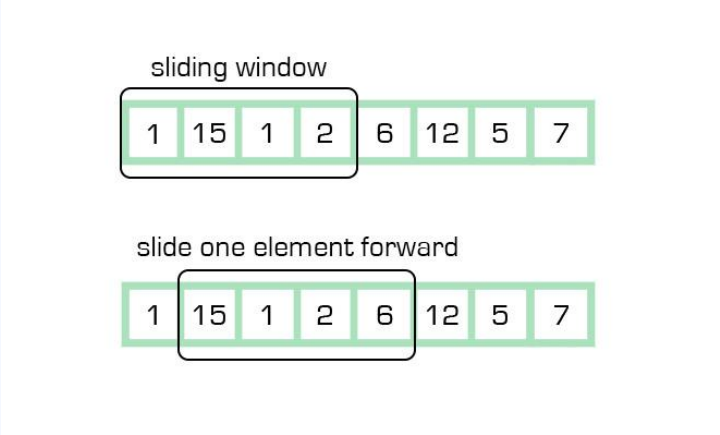
# Flow and congestion control

Preventing overwhelming the receiver
Tcp uses the sliding window mechanism
Key ideas:

- windowing
- Receive widow
- Zero window
- Window scaling
- Slow start

How TCP detects and responds to network congestion
Key ideas:

- Fast transmit
- Fast recovery
- Congestion avoidance
- Slow start



# Error Detection

How Tcp guarantees data integrity and completeness:

- Checksums
  - detect errors in transmitted data by verifying that what was sent matches what was received.
- Sequence numbers
  - label packets in order so the receiver can reassemble them correctly even if they arrive out of order.
- Acknowledgments
  -confirmations sent by the receiver to the sender to indicate that a packet was successfully received.
- Retransmission timeout
  -the time a sender waits for an acknowledgment before assuming a packet was lost and resending it.

# Retransmission

1

GO-Back-N

2

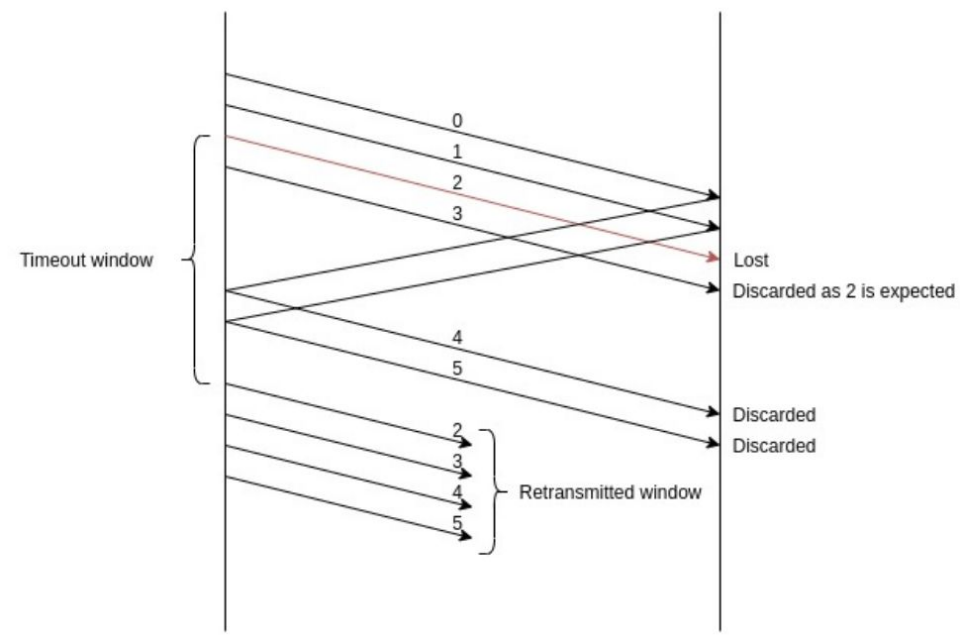
Selective Repeat

3

Stop and wait

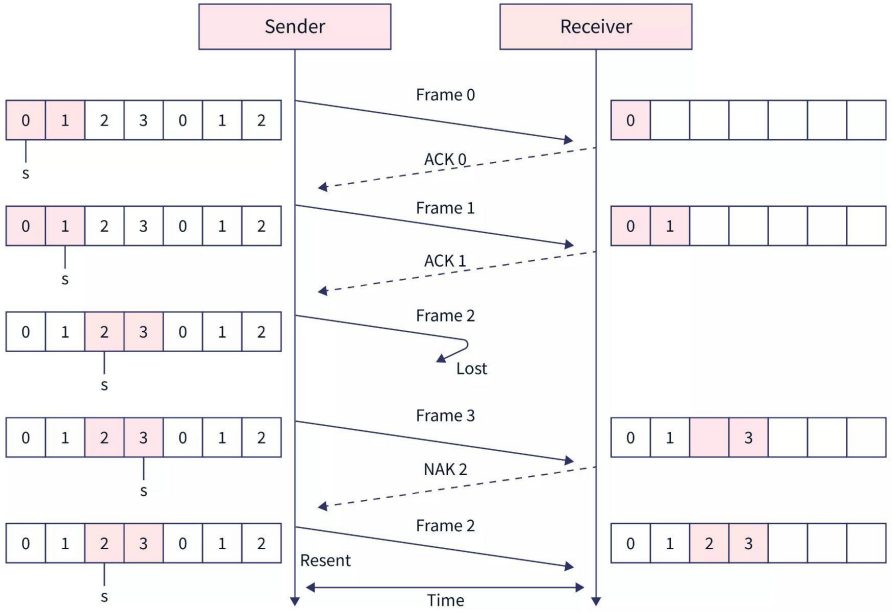
# GO-Back-N

Go back n uses a sliding window if the acknowledgement for a window isnt received it resends all of the frames in the window

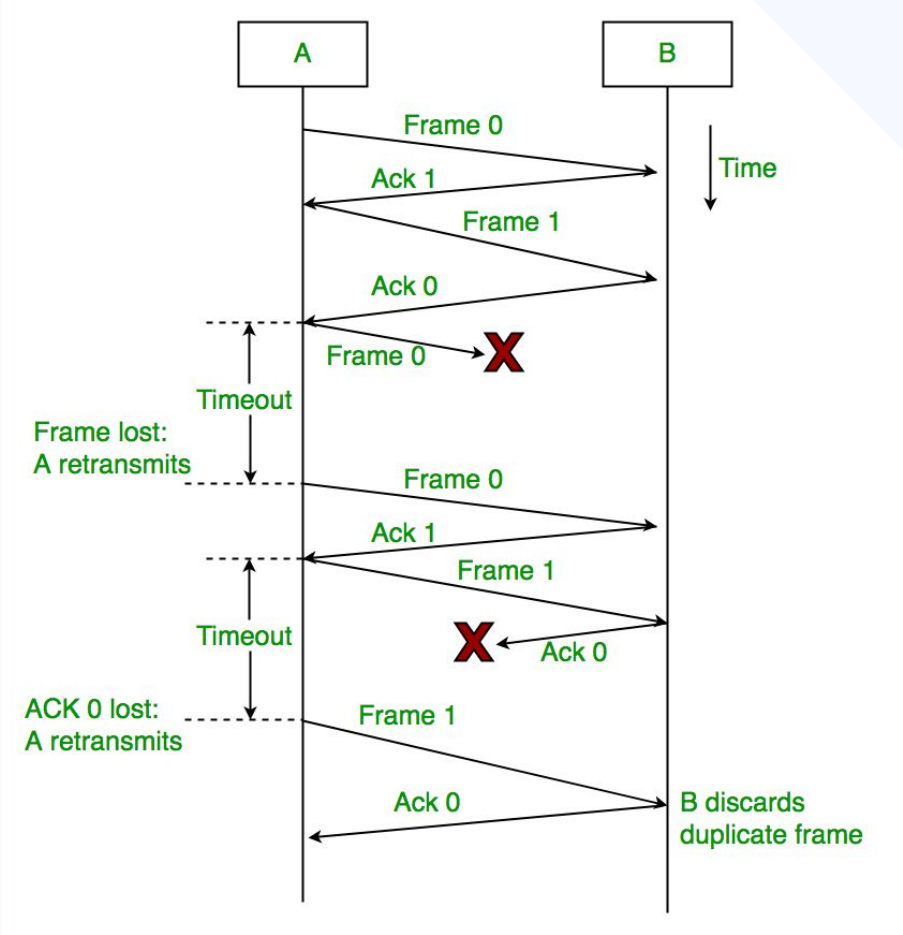


# Selective Repeat

Selective repeat  is when a frame is lost it sends only the missing frame and continues which is harder to implement then the other two




SCALER Topics

# Stop and Wait

Stop and wait happens if a frame was lost, it times out and waits then the  same frame gets retransmitted